SOUND BASED PROBLEM SOLVING PROJECT

Sound-Based Navigation System using RF Synchronization

TEAM - 25

MEMBERS

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INTRODUCTION

- Distance measurement is crucial in robotics, navigation, and automation.
- Conventional ultrasonic sensors (like HC-SR04) have limitations in range and flexibility.
- Our project introduces a sound-based navigation system that combines:
- RF synchronization - ensures precise timing reference.
- Sound-based time-of-flight (ToF) - provides actual distance measurement.
- This dual approach makes the system low-cost, accurate, and customizable.

METHODOLOGY

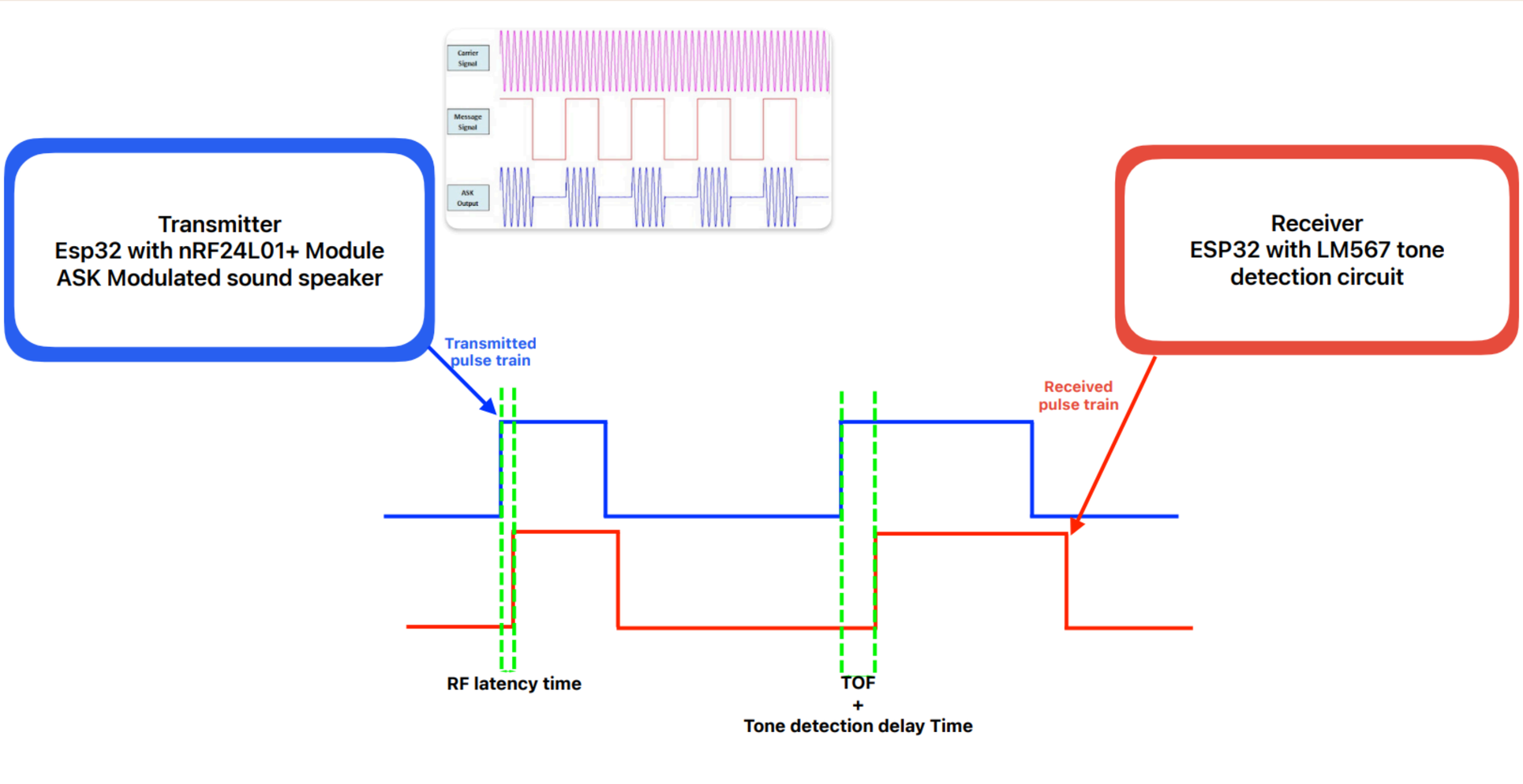
Transmitter Unit

- The ESP32 transmits a synchronization (sync) packet via the NRF24L01 RF module.
- After a fixed delay, the ESP32 generates a sound pulse.
- The sound pulse is amplitude shift keying (ASK) modulated with a 16 kHz sinusoidal carrier.
- The modulated signal is amplified using LM386 and played through a speaker.

Receiver Unit

- On receiving the RF sync, ESP32 starts a microsecond timer.
- A microphone picks up the transmitted tone.
- The signal passes into LM567 tone detector, which filters and detects only the correct frequency.
- When LM567 detects the tone, it sends a digital signal to ESP32.
- ESP32 stops the timer → calculates distance using : d = t x v

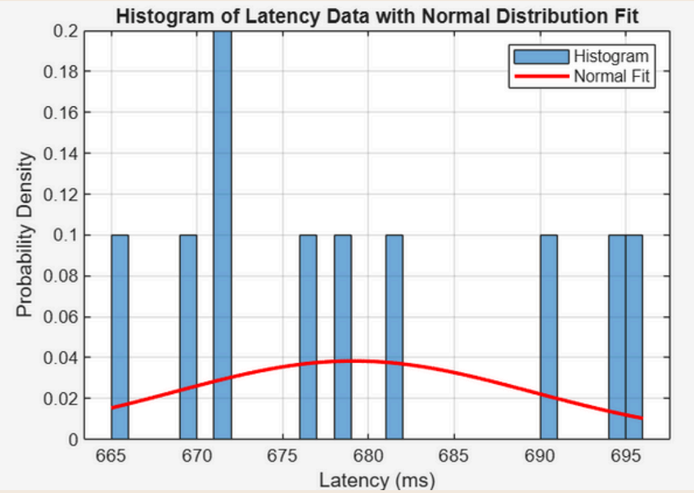
METHODOLOGY



PROBLEMS AND SOLUTIONS

RF Latency

- RF signal transmission latency can impact measurement accuracy.
- Selected NRF24L01+ module for its low latency and cost-effectiveness.
- Mean latency: 678.67 µs, standard deviation: 8.787 µs.
- Minimal latency variation ensures reliable performance.



Timer Resolution

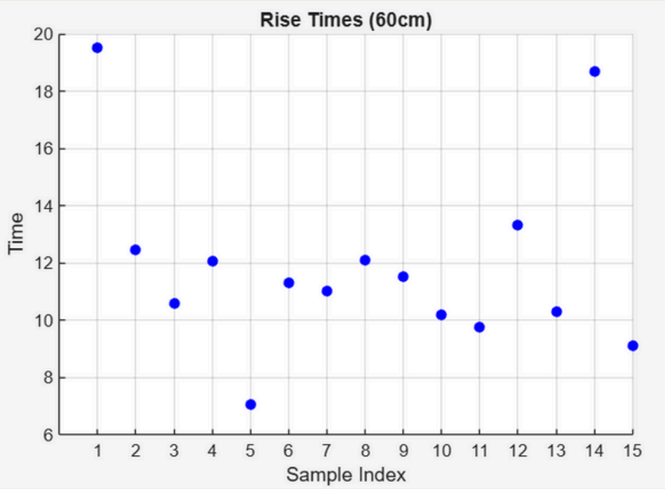
- Measurements require microsecond-level precision for accurate analysis.
- Arduino lacks the necessary accuracy and processing power for this requirement.
- ESP32 was selected for its superior precision and analysis capabilities.

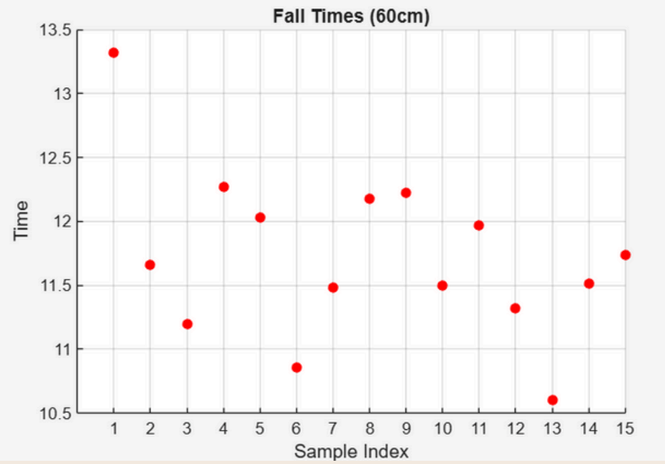
AN AUDIO SIGNAL TRAVELS APPROXIMATELY 0.343 MILLIMETERS IN 1 MICROSECOND IN AIR UNDER STANDARD CONDITIONS

TONE DETECTION

- Sound pulse generated at 16,000 Hz, detected at receiver using a microphone.
- Attempted tone detection with Goertzel algorithm on ESP32, but its ADC is limited to below 32,000 Hz.
- Used STM32 as external ADC; however, inconsistent packet capture led to highly variable delay times.
- Adopted LM567 tone detector IC for reliable tone detection.

ANALYSIS





Category	Mean	Std Dev
60cm Rise	11.93	3.17
60cm Fall	11.72	0.64
45cm Rise	13.65	6.02
45cm Fall	11.74	0.85

